

Aryan Miriyala

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PROFESSIONAL SUMMARY

AI/ML systems-oriented computer science graduate student with experience in **Python** analysis code, **C++/OpenMP** runtime experimentation, workload benchmarking, reproducible modeling, and cloud-scale data pipelines. Built performance telemetry, adaptive scheduling, ML/RL experiments, and production data workflows with emphasis on quantitative analysis, technical communication, and systems tradeoffs.

EDUCATION

Bowling Green State University <i>Master of Science in Computer Science, GPA: 4.00/4.00</i>	Bowling Green, OH <i>Aug. 2024 – Aug. 2026</i>
Bowling Green State University <i>Bachelor of Science in Computer Science, GPA: 4.00/4.00</i>	Bowling Green, OH <i>Jan. 2021 – Apr. 2024</i>

EXPERIENCE

Artificial Intelligence & Computer Science Intern <i>SmartSolve Industries</i>	May 2026 – Present <i>Bowling Green, OH</i>
- Supported AI-enabled internal software planning, including a QMS dependency register for modeling relationships between quality documents and improving document traceability. - Built secure internal software with Next.js, TypeScript, PostgreSQL, Drizzle ORM, SSO, and auth middleware for onboarding and org-chart workflows around sensitive employee data. - Created a cross-platform Docker devcontainer for WSL2/Colima, standardizing reproducible development environments for AI-assisted software development and code review.	
Data Engineering Intern <i>American Association of Insurance Services</i>	May 2025 – Aug. 2025 <i>Lisle, IL</i>
- Automated a manual SQL billing workflow with Python, PySpark, and AWS Glue, processing 20+ TB of production insurance data for 700+ member companies. - Built data validation, reverse-ETL, and custom AWS IAM access patterns across S3 and MDM workflows, improving controlled data access and reproducible production operations.	
Software Engineering Intern <i>American Association of Insurance Services</i>	May 2023 – May 2024 <i>Lisle, IL</i>
- Engineered Python/JSON backend automation generating 1,000+ production SQL tables across Oracle/PostgreSQL, supporting standardized data models across 10+ insurance lines. - Built Python/Pandas workflows with multiprocessing, fuzzy matching, and hashing to clean large CSV datasets for production data-standardization pipelines. - Automated database validation with AWS Lambda triggers that tokenized PII using hashlib/boto3 and ingested structured JSON into S3 for downstream workflows.	

PROJECTS

Self-Adaptive Parallelism <i>C++, OpenMP, Python, Docker, Linux</i> : Built a runtime research system with per-epoch telemetry and a UCB controller to tune scheduling policy, chunk size, and thread count across Mandelbrot, heat diffusion, and reduction workloads.
Performance Modeling Outputs <i>Python, CSV telemetry, benchmarking, visualization</i> : Generated reproducible runtime, speedup, efficiency, Karp-Flatt, and scheduler-decision plots comparing adaptive scheduling against serial and fixed-policy baselines.
MarioRL Agent Comparison <i>Python, PyTorch, Stable-Baselines3, Gym, TensorBoard</i> : Compared PPO and DQN agents with frame preprocessing, checkpoints, TensorBoard logs, generated clips, and reward/training-speed performance graphs.
HealthTrend Data Pipeline <i>Spark, PySpark, Kafka, Hadoop HDFS, Docker Compose</i> : Built a containerized pipeline routing CSV patient data and JSON health events into distributed storage and stream-processing workflows.

TECHNICAL SKILLS

Systems & Performance: C++, OpenMP, Python, Linux, Docker, Make, runtime instrumentation, adaptive scheduling, UCB optimization, workload characterization, throughput/latency/efficiency analysis, CSV telemetry, data visualization
AI/ML & Modeling: PyTorch, Stable-Baselines3, PPO, DQN, Gym, TensorBoard, reinforcement learning, LLM inference, RAG, prompt engineering
Data & Analysis: Pandas, PySpark, Spark, SQL, AWS Glue, S3, Lambda, IAM, Kafka, Hadoop HDFS, data validation, reproducible pipelines
Software Engineering: Git, Docker Compose, TypeScript/JavaScript, Next.js, Node.js, Express, PostgreSQL, Oracle, REST APIs, technical documentation